Switching Circuits and Logic Design (CS21202)

Module 6

Latches and Flip-Flops

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Introduction

- We shall now start our discussion on *sequential circuits*.
  - Outputs depend not only on the present inputs, but also on the past history of the system.
  - The system *memorizes* the *state* in which it is in.
- How does the circuit memorize state?
  - Using basic memory elements called *latches* and *flip-flops*.
- We shall be discussing various methods of designing latches and flip-flops.

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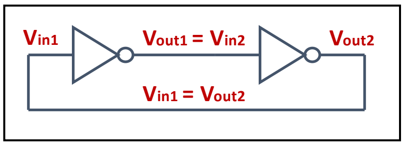
Basic Idea Behind Storing a Bit



- Consider a cascade of two inverters with feedback.
  - Constitutes a *stable state* of the system.
  - The circuit can memorize the output values as long as power is on.
- In practice, we need something more.
  - We should be able to set the output values to 0 or 1 as per our need.
  - Need some additional circuitry.
  - The exact method distinguishes between different types of *latches* and *flip-flops*.

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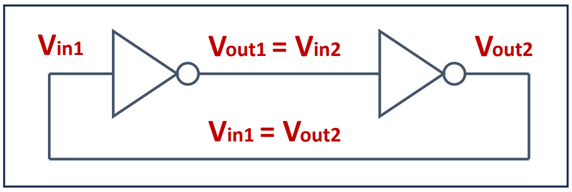

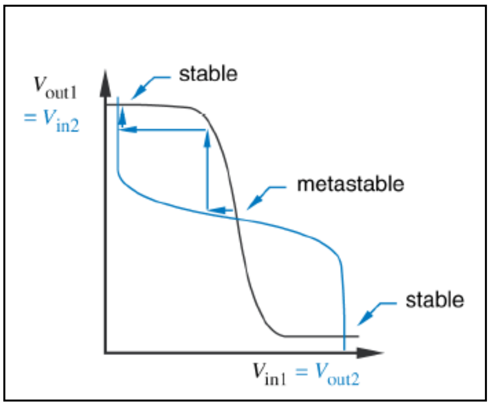

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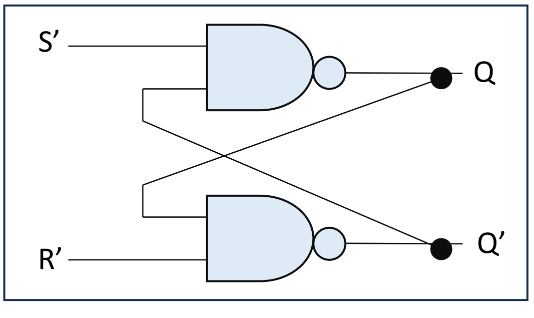
Design of Latches

- A *latch* is a temporary storage device that has two stable states, 0 and 1.
  - Level sensitive storage element (also called *bistable multivibrator*).
  - A flip-flops is a special kind of latch where a clock signal triggers the change in the stored value.
- Various types of latches / flip-flops:
  a) S-R (Set-Reset) type
  b) D (Delay) type
  c) J-K type
  d) T (Toggle) type

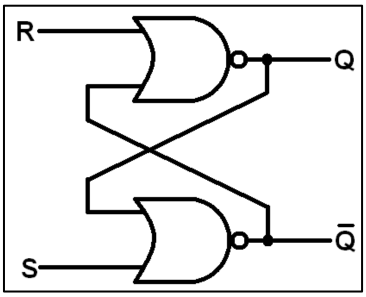
(a) The Set-Reset (S-R) Latch

- Consists of a pair of cross-coupled NAND or NOR gates.
  - Two inputs (*S* and *R*) and two outputs (*Q* and *Q'*).
  - The output can be set to 0 or 1 by applying suitable values on *S* and *R* inputs.



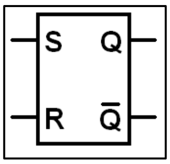
S	R	Q_{n+1}	Q'_{n+1}
0	0	Q_n	Q'_n
0	1	0	1
1	0	1	0
1	1	?	?

- Alternate realization using NOR gates:

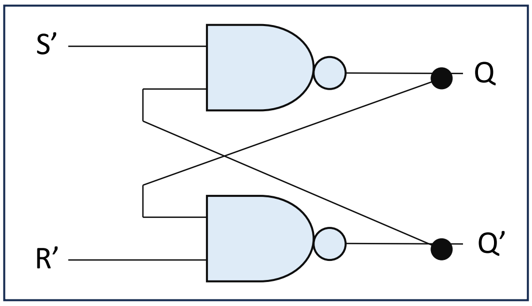


S	R	Q_{n+1}	Q'_{n+1}
0	0	Q_n	Q'_n
0	1	0	1
1	0	1	0
1	1	?	?

Why is S = R = 1 an invalid input?



Logic symbol for S-R latch

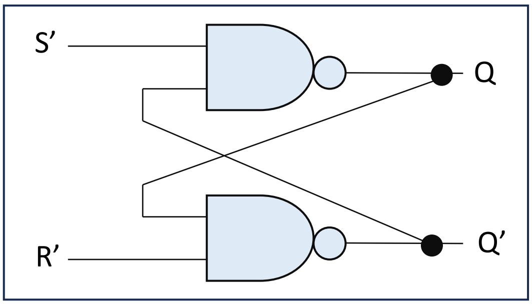


S	R	Q_{n+1}	Q'_{n+1}
0	0	Q_n	Q'_n
0	1	0	1
1	0	1	0
1	1	?	?

After applying S = R = 1, both Q and Q' becomes 1. But if we apply S = R = 0 after that, Q will be either 1 or 0 depending on the relative delays of the two gates (called *race condition*).

Race condition:

- A scenario where the final output depends on the relative speeds of various components (here gates).
- If we apply S = R = 1, and then apply S = R = 0, the outputs will settle to either Q = 0, Q' = 1 or Q = 1, Q' = 0 depending on the relative speeds of the two gates.



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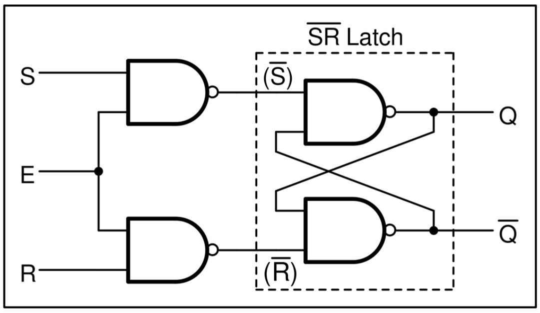
Gated S-R Latch

- It is a S-R latch with an *enable* input (*E*).
  - When *E* = 1, the latch is *active*.
  - When *E* = 0, the latch is *de-active* and the outputs do not change.



E	S	R	Q_{n+1}	Q'_{n+1}
0	X	X	Q_n	Q'_n
1	0	0	Q_n	Q'_n
1	0	1	0	1
1	1	0	1	0
1	1	1	?	?

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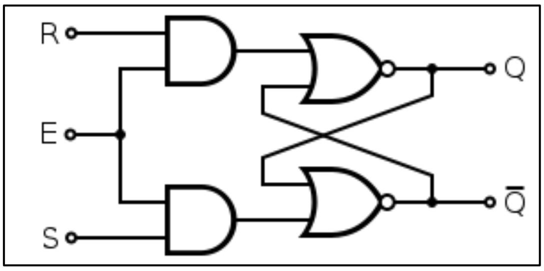
Gated S-R Latch :: Alternate Design



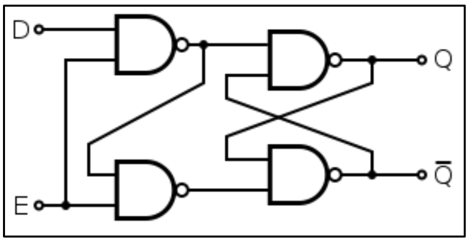
E	S	R	Q_{n+1}	Q'_{n+1}
0	X	X	Q_n	Q'_n
1	0	0	Q_n	Q'_n
1	0	1	0	1
1	1	0	1	0
1	1	1	?	?

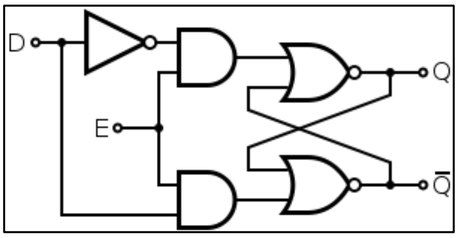
(b) The D Latch

E	D	Q_{n+1}	Q'_{n+1}
0	X	Q_n	Q'_n
1	0	0	1
1	1	1	0

- A D-latch has a single data input *D* and an enable input
  - Sometimes referred to as *Gated D-latch*.
- When the enable input *E* is active (high), the output *Q* will follow the input *D*.
  - Also referred to as a *transparent latch*.
- When the enable input *E* is inactive (low), the latch holds the last value of *D* and the output remains unchanged.

- Realization of *gated D-latch* using gates:





E	D	Q_{n+1}	Q'_{n+1}
0	X	Q_n	Q'_n
1	0	0	1
1	1	1	0

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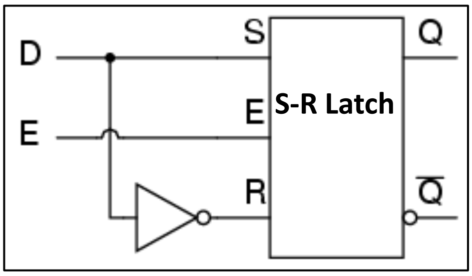
Realization of a D latch using S-R latch



D latch is most widely used in real circuits.

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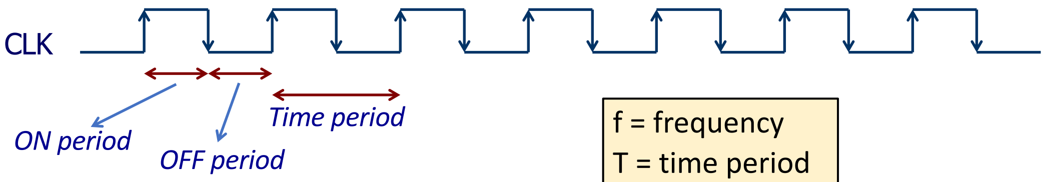
Notion of Clock

- A *clock* is a periodic (repetitive) rectangular pulse train that synchronizes the operation of a synchronous sequential circuit.
- For synchronous operation, the storage elements must be triggered by the clock (*rising edge* or *falling edge*).
  - "*posedge*" and "*negedge*" in Verilog.
  - Such storage elements are called *flip-flops* (as opposed to latches, which are *level triggered*).

Edge-Triggered Flip-Flop

- An *edge-triggered flip-flop* changes its state in synchronism with a clock pulse.
  - Either at the *positive-edge* or at the *negative-edge*.
  - Useful in the design of *synchronous sequential circuits*, where all changes in the circuit outputs occur in synchronism with the clock.



f = frequency
T = time period
T = 1 / f

Other Terminologies Related to Clock

- **Duty Cycle**: Ratio of ON period to time period, often expressed as a percentage.
- **Clock Skew**: The active edge of the clock reaches various points at unequal times.
- **Clock Jitter**: Variations in the time at which the active edge of the clock reaches a node.

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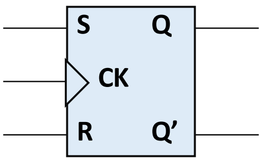
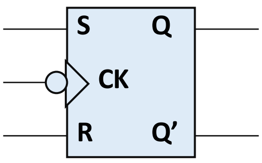
Edge-Triggered S-R Flip-Flop



Positive edge triggered



Negative edge triggered

Positive edge triggered state table

CK	S	R	Q_{n+1}	Q'_{n+1}
0/1	X	X	Q_n	Q'_n
↑	0	0	Q_n	Q'_n
↑	0	1	0	1
↑	1	0	1	0
↑	1	1	?	?

Q(t) \ SR	00	01	11	10
0	0	0	X	1
1	1	0	X	1

Characteristic equation:

R.S = 0

Q(t+1) = R'.Q(t) + S

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Excitation Table

- Consider a flip-flop (any type) with output *Q*.
- The excitation table shows the inputs that need to be applied to make the output *Q* go through any arbitrary transition.

  0 → 0

  0 → 1

  1 → 0

  1 → 1

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- *Excitation table* for S-R flip-flop:
  - It specifies the *S* and *R* inputs that are required to make all possible output transitions.

State Table

S	R	Q_{n+1}	Q'_{n+1}
0	0	Q_n	Q'_n
0	1	0	1
1	0	1	0
1	1	?	?

Excitation Table

Circuit changes		Required value	
From	***To***	***S***	***R***
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

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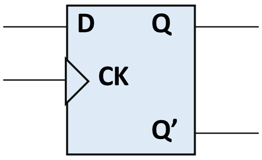
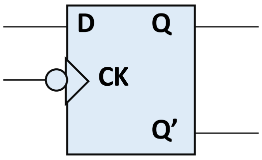
Edge-Triggered D Flip-Flop



Positive edge triggered



Negative edge triggered

Positive edge triggered state table

CK	D	Q_{n+1}	Q'_{n+1}
0/1	X	Q_n	Q'_n
↑	0	0	1
↑	1	1	0

Characteristic equation:

$Q(t+1) = D$

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- *Excitation table* for D flip-flop:

State Table

E	D	Q_{n+1}	Q'_{n+1}
0	X	Q_n	Q'_n
1	0	0	1
1	1	1	0

Excitation Table

Circuit changes		Required value
From	***To***	***D***
0	0	0
0	1	1
1	0	0
1	1	1

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(c) J-K Flip-Flop

- It is a more general form of the S-R flip-flop where the problem of invalid input combination of $S = 0$, $R = 0$ is eliminated.
  - When we apply $J = 1$ and $K = 1$, the output Q toggles (changes from 0 to 1, or from 1 to 0.
  - Behaves similar to S-R flip-flop for all other input combinations.
- A J-K storage element cannot be implemented as a latch.
  - The output will continuously toggle if we apply $J = 1$ and $K = 1$.
  - Hence, we realize it as *J-K flip-flop*.

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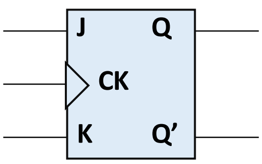
Edge-Triggered J-K Flip-Flop

- The J-K flip-flop is the most versatile flip-flop.
  - Like in S-R flip-flop, it has two inputs J and K, but does not have any invalid input combination.



Positive edge triggered

Positive edge triggered state table

CK	J	K	Q_{n+1}	Q'_{n+1}
0/1	X	X	Q_n	Q'_n
↑	0	0	Q_n	Q'_n
↑	0	1	0	1
↑	1	0	1	0
↑	1	1	Q'_n	Q_n

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State Table

CK	J	K	Q_{n+1}	Q'_{n+1}
0/1	X	X	Q_n	Q'_n
↑	0	0	Q_n	Q'_n
↑	0	1	0	1
↑	1	0	1	0
↑	1	1	Q'_n	Q_n

Excitation Table

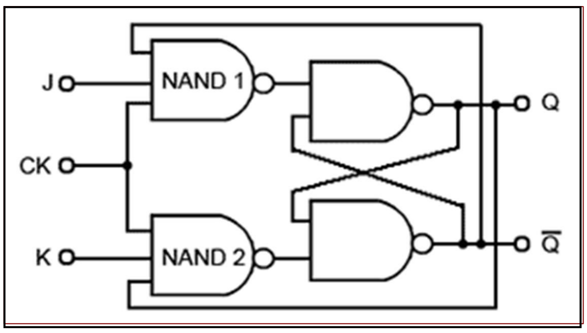
Circuit changes		Required value	
From	*To*	*J*	*K*
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Q(t) \ JK	00	01	11	10
0	0	0	1	1
1	1	0	0	1

Characteristic equation:

Q(t+1) = J.Q(t)’ + K’.Q(t)

Gate-Level Implementation of J-K Flip-Flop



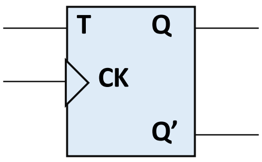
Assume that a single output transition will occur for J = K = 1.

CK acts like an enable input; edge detection circuit is not shown (but must be present for J-K flip-flop).

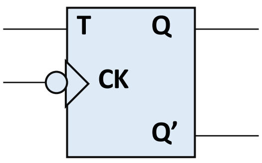
(d) T flip-flop

- The T flip-flop is a simplified version of the J-K flip-flop.
  - It has a single input T (*toggle*).
  - The output Q changes state (toggles) whenever $T = 1$ and a clock pulse is applied.
  - When $T = 0$, the flip-flop holds its present state and the output Q does not change.

Edge-Triggered T Flip-Flop



Positive edge triggered



Negative edge triggered

Positive edge triggered state table

CK	T	Q_{n+1}	Q'_{n+1}
0/1	X	Q_n	Q'_n
↑	0	Q_n	Q'_n
↑	1	Q'_n	Q_n

Characteristic equation:

$$Q(t+1) = T \oplus Q(t)$$

State Table

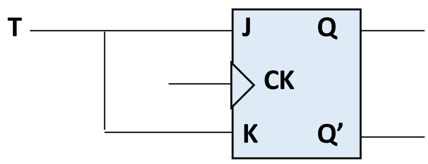
CK	T	Q_{n+1}	Q'_{n+1}
0/1	X	Q_n	Q'_n
↑	0	Q_n	Q'_n
↑	1	Q'_n	Q_n

Excitation Table

Circuit changes		Required value
From	***To***	***T***
0	0	0
0	1	1
1	0	1
1	1	0

Implementing a T flip-flop using J-K flip-flop

- Use only the two input combinations $J = K = 0$, and $J = K = 1$.



T flip-flop can divide the frequency of the clock by 2

- Apply *T* = 1, and apply a rectangular waveform at the *CLOCK* input (frequency *f*, arbitrary duty cycle).
- At the output *Q*, we shall obtain a square wave with a 50% duty cycle and a frequency of 2*f*.
- Show the timing diagram.

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Flip-flops with Asynchronous Preset and Clear

- It is often necessary to initialize a flip-flop to a known state before the circuit operation starts.
  - *Preset* (sets *Q* = 1) and *Clear* (sets *Q* = 0).
  - These are asynchronous operations, and can be easily implemented by using additional inputs on the cross-coupled gates.
    - The term "*asynchronous*" means that it does not depend on the clock.
  - Synchronous versions also exist, where the preset and clear operations occur in synchronism with the clock.

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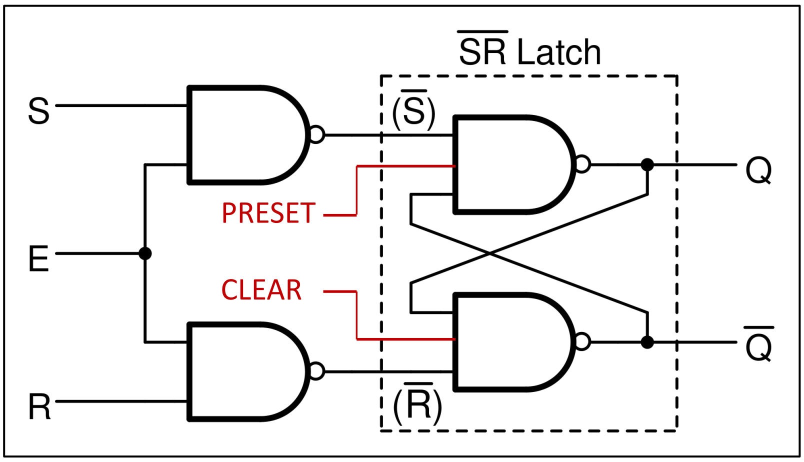

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How is Edge Triggering Implemented?

- Why is edge triggering required?
  - For latches with *ENABLE* input, the latch remains open as long as the *ENABLE* input is active (if inputs change, outputs also change).
  - Not desirable in many applications, where flip-flop outputs drive other circuits.
  - One solution is to apply a very narrow *ENABLE* signal so that the circuit changes state only once.

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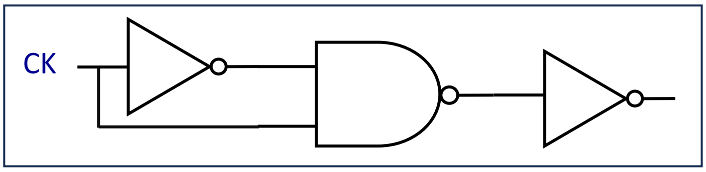

Narrow ENABLE pulse

A simple solution:
- Width of the pulse depends on the delay of the NOT gate.
- We can use any odd number of NOT gates.

Drawback:

Delay depends on the circuit

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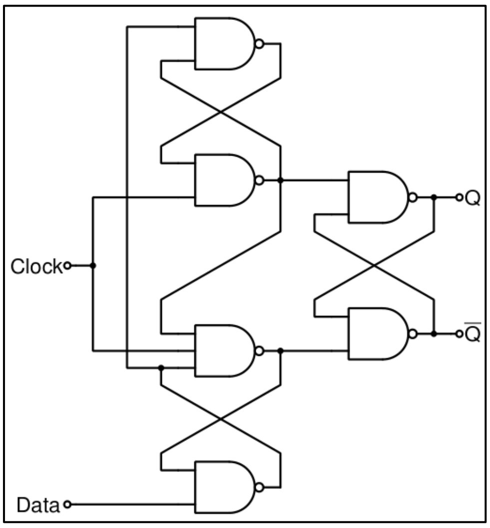

A positive-edge triggered D flip-flop.
- Remembers the clock edge in the cross-coupled latch in the first stage.

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Master-Slave Flip-Flops

- A master-slave flip-flop has two latch stages, called *Master* and *Slave*.
- Data gets stored in the *Master* stage one one edge or level of a control signal (*clock*), and is transferred to the *Slave* on the other edge or level of the control signal.
- Aims to address the problem in latches when the *ENABLE* signal may cause multiple changes in the storage cell (for J-K and T types).
- Suffers from a problem called 0*'s catching* and 1*'s catching* that may be undesirable for many applications.
  - Edge-triggered versions are preferred in such cases.

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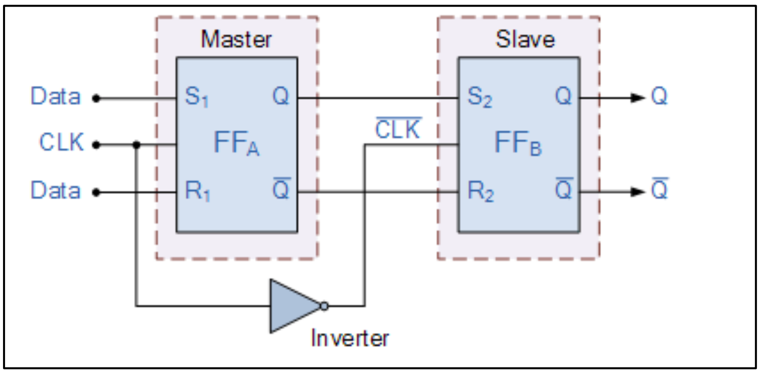
(a) S-R master-slave flip-flop



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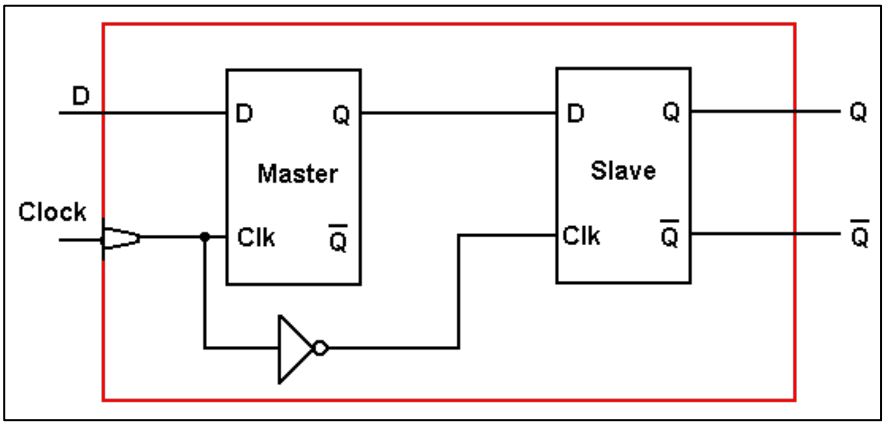
(b) D master-slave flip-flop



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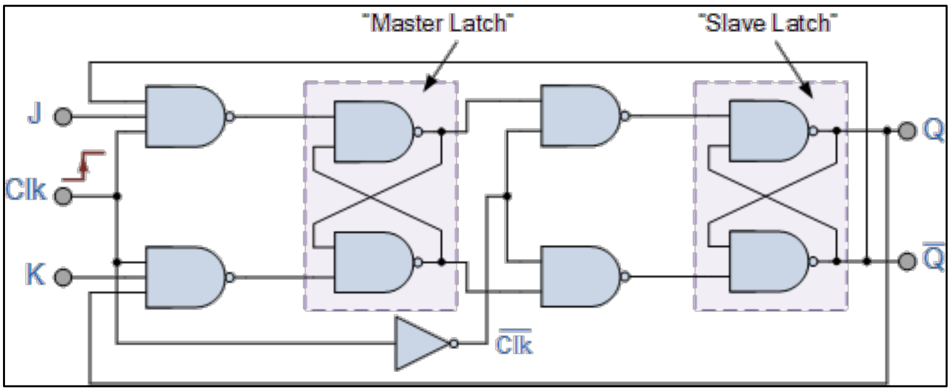
(c) J-K master-slave flip-flop



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Converting One Flip-Flop Type to Another

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Basic Steps of Conversion

Suppose we want to convert a flip-flop *X* to flip-flop *Y*.

1. Start with the state table and the excitation table of flip-flop *X*.
2. From the state table, form a *conversion table* that shows the required transitions for flip-flop *Y*.
3. Minimize the functions for the inputs corresponding to flip-flop *Y*, and arrive at the final realization.

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Example 1: S-R flip-flop to D flip-flop

Conversion Table

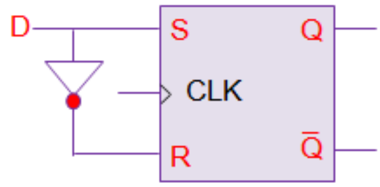
D	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

D \ Q_n	0	1
0		
1	1	X

S = D

D \ Q_n	0	1
0	X	1
1		

R = D’



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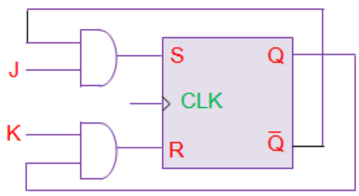
Example 2: S-R flip-flop to J-K flip-flop

Conversion Table

J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1



J \ KQ_n	00	01	11	10
0		X		
1	1	X		1

$S = J.Q_n’$

J \ KQ_n	00	01	11	10
0	X		1	X
1			1	

$R = K.Q_n$

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Example 3: S-R flip-flop to T flip-flop

Conversion Table

T	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	X	0
1	0	1	1	0
1	1	0	0	1

T \ Q_n	0	1
0		X
1	1	

$S = T.Q_n'$

T \ Q_n	0	1
0	X	
1		1

$R = T.Q_n$

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Example 4: J-K flip-flop to S-R flip-flop

Conversion Table

S	R	Q_n	Q_{n+1}	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	?	?	X	X

S \ RQ_n	00	01	11	10
0		X	X	
1	1	X	X	X

J = S

S \ RQ_n	00	01	11	10
0	X		1	X
1	X		X	X

K = R

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Example 5: J-K flip-flop to D flip-flop

Conversion Table

D	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

D \ Q_n	0	1
0		X
1	1	X

J = D

D \ Q_n	0	1
0	X	1
1	X	

J = D'

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Example 6: J-K flip-flop to T flip-flop

Conversion Table

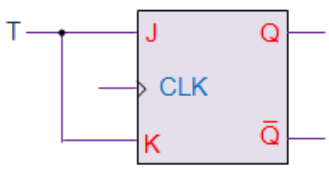
T	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

T \ Q_n	0	1
0		X
1	1	X

J = D

T \ Q_n	0	1
0	X	
1	X	1

K = T



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Example 7: D flip-flop to S-R flip-flop

Conversion Table

S	R	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	?	?	X

S \ RQ_n	00	01	11	10
0				
1				

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Example 8: D flip-flop to J-K flip-flop

Conversion Table

J	K	Q_n	Q_{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	1

J \ KQ_n	00	01	11	10
0				
1				

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Example 9: D flip-flop to T flip-flop

Conversion Table

T	Q_n	Q_{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

D \ Q_n	0	1
0		
1		

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Example 10: T flip-flop to S-R flip-flop

Conversion Table

S	R	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	?	?	X

S \ RQ_n	00	01	11	10
0				
1				

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Example 11: T flip-flop to J-K flip-flop

Conversion Table

J	K	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

J \ KQ_n	00	01	11	10
0				
1				

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Example 12: T flip-flop to D flip-flop

Conversion Table

D	Q_n	Q_{n+1}	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

D \ Q_n	0	1
0		
1		

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Clocking and Timing

Synchronous Sequential Circuit

- A sequential circuit is a digital circuit that contains storage elements, such as latches and flip-flops.
- In a synchronous sequential circuit, all operations are synchronized by a clock signal.
  - We use flip-flops.
  - Some factors determine the maximum clock frequency at which the circuit can operate correctly.

Timing Issues

- For the correct operation of a synchronous sequential circuit, several timing-related issues must be considered.
  - Some of these issues relate to the *properties of flip-flops*.
  - Some of these issues relate to *circuits external to the flip-flops*.
- All these taken together determine the maximum speed with which the circuit can operate.
  - Typically specified in terms of the clock frequency.

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Some Definitions

- **Clock width (t_w):** Minimum time duration for which the clock signal needs to be high in order that the flip-flops it feeds work properly.
- **Setup time (t_{setup}):** Amount of time the input to a flip-flop must be stable *before* the clock transitions high (for positive-edge triggered), or transitions low (for negative-edge triggered).
- **Hold time (t_{hold}):** Amount of time the input to a flip-flop must be stable *after* the clock transitions high (for positive-edge triggered), or transitions low (for negative-edge triggered).

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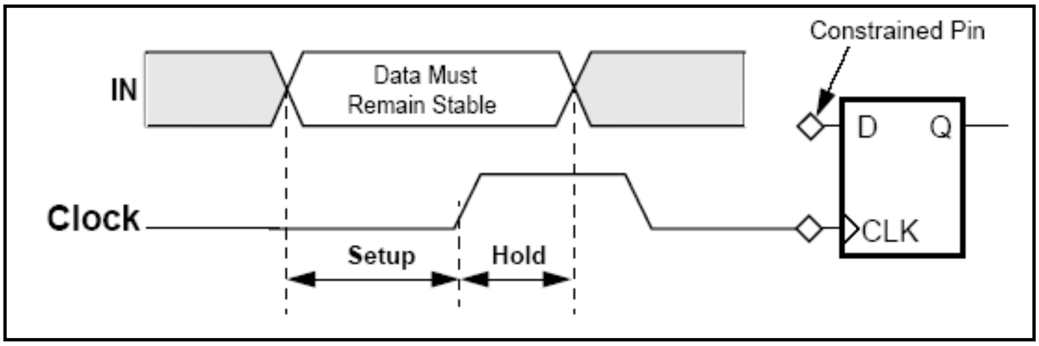
- **Propagation delays ($t_{p\text{-}lh}$ and $t_{p\text{-}hl}$):** Delay between clocking event (low-to-high or high-to-low transition) and change in the output.



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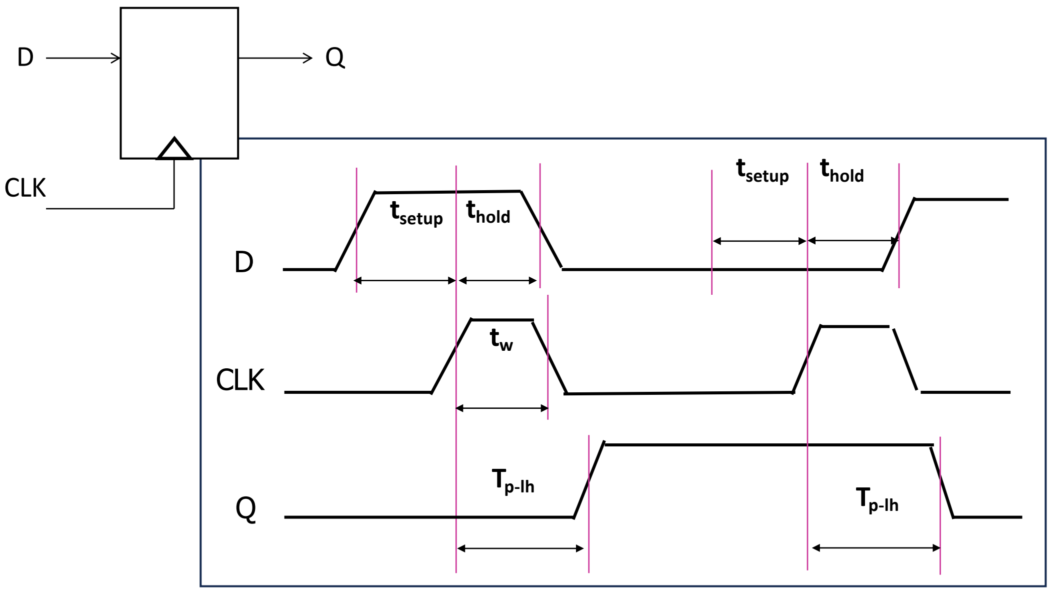
D flip-flop



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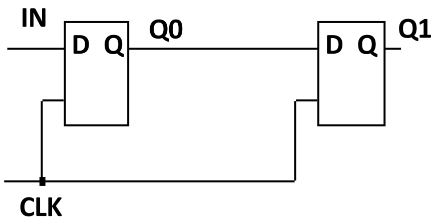
Cascading Flip-flops

- Suppose that the flip-flop propagation delay exceeds the hold time.
- Second stage can commit its input before Q0 changes.



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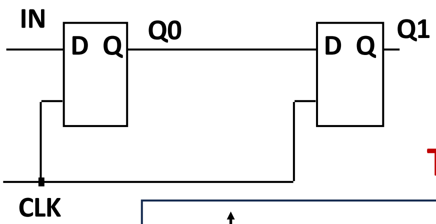
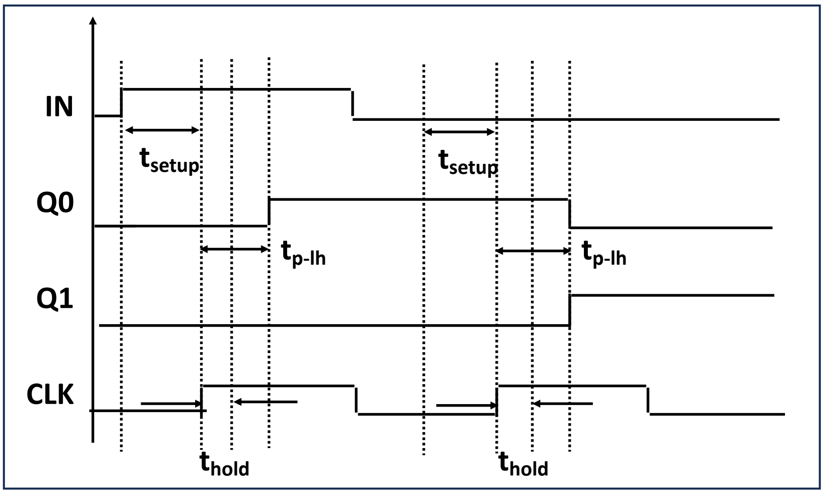


The correct scenario



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Edge-triggered Clocking

- Edge-triggered operation (either positive or negative edge triggered)
  - Data leaving at time *t* must arrive at the next flip-flop *one setup time before t + T*, where *T* is the clock period.
- Clocking relationships are relatively simple:
  - Delay from each input flip-flop to each output flip-flop of a combinational block should be less than $(T - t_{setup})$.
- Ideally, the clock signal should arrive at *all* flip-flops *simultaneously*.
  - In practice, *clock skew* exists.

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Example 1

- Given a clock signal of frequency f Hz, how to generate another clock of frequency f/2 Hz.
  - Frequency division.
  - Already discussed – use a T flip-flop.

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Example 2

- Given a clock signal of frequency *f* Hz, how to generate another clock of frequency $f/2^k$ Hz, for some integer *k*.
    - Frequency division by some power of 2.
    - Use *k* T flip-flops in cascade, with the *Q* output of one flip-flop connected to the *CLK* input of the next.
    - This is the basic principle of designing a binary (ripple) counter.

Hazards

Static and Dynamic Hazards

- Hazards represent a momentary transition to the opposite logic value at the output of a circuit.
- Two types of hazards are possible:
  - *Static Hazard*: Indicates a momentary transition when the initial and final values are the same.
    - Examples: 0 → 1 → 0, or 1 → 0 → 1
  - *Dynamic Hazard*: Indicates a momentary transition when the initial and final values are different.
    - Examples: 0 → 1 → 0 → 1, or 1 → 0 → 1 → 0

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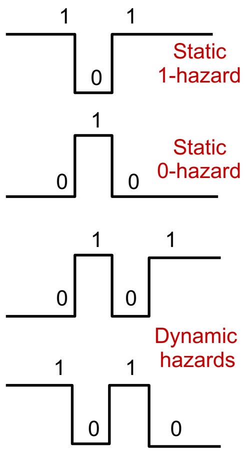
Kinds of Hazards



Input change causes output to go from 1 → 0 → 1

Input change causes output to go from 0 → 1 → 0

Input change causes a double change:
from 0 → 1 → 0 → 1 OR
from 1 → 0 → 1 → 0

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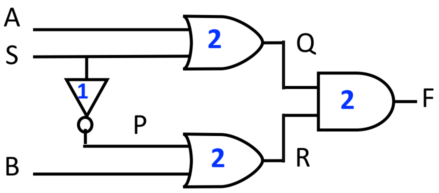
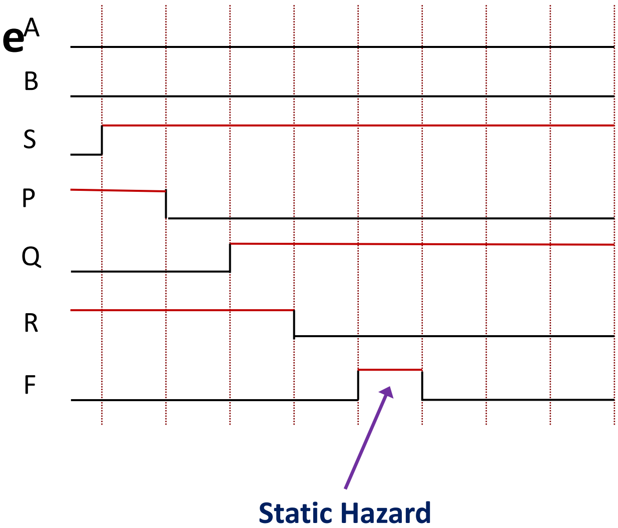
Static Hazard Example





Delay of NOT = 1
Delay of AND, OR = 2

<ABS> changes from 000 to 001

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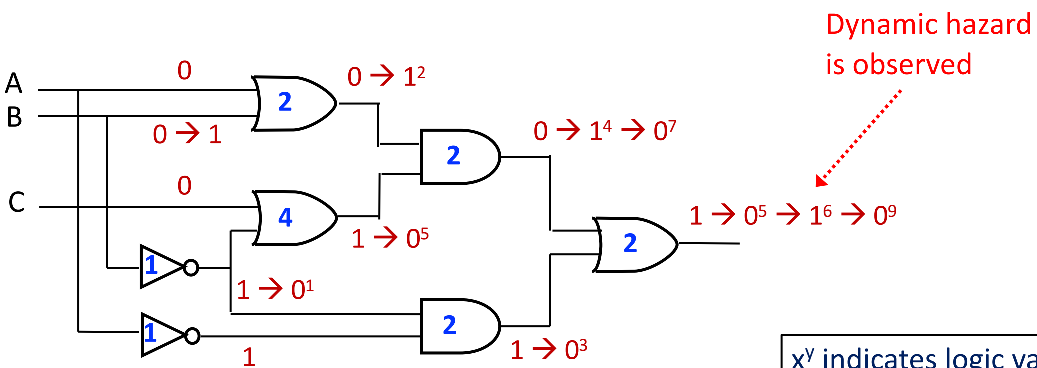
Dynamic Hazard Example



x^y indicates logic value x at time y

<ABC> changes from 000 to 010

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